PREDICTIVE ANALYTICS

Prediction models on the customer's own data, from a catalogue of five use-case families, on the platform the customer uses

Start with a Pilot on one use case, measure the business value against a control group, then move the model to production and run it

1 **Why prediction**
Business questions, what the package is, lifecycle and platforms. p. 2 – 4

2 **Five use-case families**
Retention, demand, pricing, risk, industrial operations. p. 5 – 9

3 **Results and value**
Results from the field, how business value is measured. p. 10 – 11

4 **Working together**
Three tiers, the Pilot, data platform, next step. p. 12 – 15

Business Questions That Need a Prediction

KOMTAS

Reports show what has already happened. The questions below are about what will happen next, and they come up in banking, retail, manufacturing and subscription businesses.

01 · WHY PREDICTION

«AX DOĞRULASIN»

Which customers will leave?

- Cancellations come without warning
- Balances or segments fall before the exit
- Retention offers go to everyone, or too late

Retention & Churn

How much will be needed, and where?

- Orders are based on last year's figures
- Stock-outs and overstock at the same time
- Promotions and holidays are planned by hand

Demand & Supply

Which offer or price, for whom?

- The same offer goes to every customer
- Prices and markdowns follow fixed rules
- Cross-sell depends on staff experience

Pricing & Personalization

Who will not pay?

- Every late payer is contacted the same way
- Credit decisions use a few variables
- Limits are not reviewed per customer

Risk & Collections

Where do energy and yield go?

- Set points follow operator habit
- Quality checks are done by hand
- Failures are found after they happen

Industrial Operations

Results are hard to prove

- Model projects end as a presentation
- No control group, so no measured effect
- Business teams do not use the output

Measured against a control group

WHAT A PREDICTION ADDS

A score per customer
churn, risk or propensity

A forecast per series
product, store, site or period

A price or an offer
per customer or per product

A setting per process
energy, yield, quality

A measured effect
against a control group

What Predictive Analytics Is

KOMTAS

A service package that builds prediction models on the customer's own data, puts them into daily use and measures their value. One use case at a time, from five families.

02 · WHAT IT IS

FIVE USE-CASE FAMILIES

«AX DOĞRULASIN»

Retention & Churn
Path to Churn, balance churners, segment downgrade, subscription churn

Demand & Supply
demand forecasting, inventory productivity, visitor and sales prediction

Pricing & Personalization
next best offer, recommendation, pricing, markdown, segmentation

Risk & Collections
collection optimisation, credit risk models, client limits

Industrial Operations
energy optimisation, predictive quality, yield, maintenance

FOR WHOM

Banks, retailers, manufacturers and subscription businesses with several years of customer, sales or process data.

On the platform the customer uses
Google Cloud, Dataiku, Azure Machine Learning or an open-source stack. The lifecycle and the tiers are the same on each. One use case is proven first in a Pilot.

WHEN IT IS USEFUL

«AX DOĞRULASIN»

- Churn or segment downgrade is rising
- Stock-outs and overstock at the same time
- Prices, offers or limits are set by fixed rules
- Delinquency or collection cost is rising
- Energy use or yield loss in production is high

WHAT IT IS NOT

«AX DOĞRULASIN»

- Data or analytics platform migration (separate service)
- Cleansing data in source systems
- Running the business actions, or guaranteeing their result
- Generative AI and LLM assistants (separate package)
- Real-time (online) scoring, unless agreed

Lifecycle and Delivery Platforms

KOMTAS

Each use case follows the same six steps on any platform. The first four form the Pilot; deployment and monitoring follow when the Pilot result is accepted.

03 · LIFECYCLE

«AX DOĞRULASIN»

1 Data readiness check

History, quality and the label: is the data enough for this use case?

2 Feature set

Variables built from the raw data, agreed with the business team.

3 Model

Training and selection of the model on historical data.

4 Validation

Technical metrics on hold-out data; design of the value test.

5 Deployment

Scheduled scoring, model registry, output to business systems.

6 Monitoring

Drift and performance tracked; retraining on a schedule.

A PILOT

B PRODUCTION

C RUN

MODEL ARCHITECTURE · THE SAME ON EVERY PLATFORM

Sources
CRM, core systems, ERP, sensors, web

→ **Data platform**
data and feature tables

→ **Training and registry**
training, model registry, pipelines

→ **Output tables**
scores, forecasts, set points

→ **Business systems**
CRM, campaign, ERP, collection, plant

→ **Monitoring**
drift, performance, reports

DELIVERY PLATFORM OPTIONS · AGREED IN THE SCOPE

«AX DOĞRULASIN»

Google Cloud

BigQuery, BigQuery ML and Vertex AI for training, registry and batch prediction. «GÜNER DOĞRULASIN»

Dataiku

Used in most of the reference projects, on premises or in the cloud.

Azure Machine Learning

With the customer's Azure data platform.

Open-source stack

Python libraries and open-source MLOps tools.

Family 1: Retention & Churn

KOMTAS

Which customers are likely to leave or reduce their business, and why. The output is used to act before the decision is made.

04 · RETENTION & CHURN

THE QUESTION

Who is likely to leave, and why?

A score per customer for churn, balance churn or segment downgrade, with the main drivers behind it.

USE CASES IN THIS FAMILY

Path to Churn
the steps before the decision to leave

'Balance' churners
funds withdrawn soon after they arrive

Asset churn / segment downgrade
bank customers moving to a lower segment

Subscription churn
members who cancel or do not renew

Customer satisfaction index
a measure of experience beyond NPS

TYPICAL DATA «AX DOĞRULASIN»

- At least 12–24 months of customer history
- Customer, product, membership or contract data
- Transactions, balances, visits or usage
- Campaigns, contacts and complaints
- An agreed definition of churn or downgrade

HOW SUCCESS IS MEASURED

Technical metric
AUC or Gini; lift in the top decile on hold-out data

Business metric
Retention or downgrade difference, test group vs control group

REFERENCES AND RESULTS «AX DOĞRULASIN»

Odeabank · asset churn prediction
11% lower segment downgrade in the test group vs the control group

A fitness / subscription company · subscription churn
Result: «RESULT» «GÜNER DOĞRULASIN»

Family 2: Demand & Supply

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How much will be sold or needed, per product, store, site and period. The forecast feeds ordering, stock, staffing and lease decisions.

05 · DEMAND & SUPPLY

THE QUESTION

How much will be needed, where and when?

A forecast for each series (for example product–store or store–day) and each period in the planning horizon.

USE CASES IN THIS FAMILY

Demand forecasting
retail and manufacturing

Inventory productivity
overstock found against realistic targets

Visitor and sales prediction
shopping malls and store turnover

Schedule optimisation
production or staff plans built on the forecast

TYPICAL DATA «AX DOĞRULASIN»

- At least 2–3 years of sales history
- Sales by product, store or site, by day or week
- Price, promotion and stock-out history
- Calendar, holidays and weather
- External indicators: inflation, market, card spending

HOW SUCCESS IS MEASURED

Technical metric
WAPE / MAPE and bias against the current method on a back-test period

Business metric
Stock holding cost, stock-outs and overstock

REFERENCES AND RESULTS «AX DOĞRULASIN»

A tyre manufacturer · demand prediction
7% lower stock holding cost for 7 tyre types

Shopping mall management · visitor and sales prediction
Forecasts for 350+ stores in 2 malls, 1.6M visitors

A retail company · demand forecasting
Result: «RESULT» «GÜNER DOĞRULASIN»

Family 3: Pricing & Personalization

KOMTAS

Which offer, product or price for each customer or product group. The output is used in offers, price lists, campaigns and contact rules.

06 · PRICING & PERSONALIZATION

THE QUESTION

Which offer, product or price, for whom?

Propensity, value and price response per customer or product, used in offers, price lists and contact rules.

USE CASES IN THIS FAMILY

- **Next best offer** — propensity to buy the next product
- **Product bundling** — products that sell together
- **Recommendation** — top-N products per customer or context
- **Customer lifetime value** — long-term value of each customer
- **Deposit pricing** — rate decisions per customer (banking)
- **Credit pricing** — pricing at a finer level per applicant
- **List price and markdown** — price elasticity per segment
- **Dynamic pricing** — prices by time, capacity and use
- **Customer segmentation** — groups with similar needs
- **Customer contact policy** — channel and timing of contact
- **Onboarding analytics** — the first months of new customers

TYPICAL DATA — «AX DOĞRULASIN»

- Sales, price, discount and negotiation history
- Customer and product segments, product hierarchy
- Views, baskets and purchases (at least 6–12 months)
- Offers, campaigns, contacts and responses
- For banks: rates, renewals and competitor prices

HOW SUCCESS IS MEASURED

Technical metric
Elasticity fit, propensity lift, precision@k and recall@k

Business metric
A/B test: revenue, conversion, click-through rate

REFERENCES AND RESULTS — «AX DOĞRULASIN»

A spare-parts seller · list price and markdown optimisation
+4% revenue at the same conversion in the test group; price elasticity for 30,000+ SKUs; 12 A/B test scenarios

Family 4: Risk & Collections

KOMTAS

Which customers will have difficulty paying, and which action or limit fits each of them. The output is used in credit decisions and in the collection process.

07 · RISK & COLLECTIONS

THE QUESTION

Who will not pay, and what action works?

Risk scores for applicants, limit holders and late payers, and the contact or limit decision that follows.

USE CASES IN THIS FAMILY

Collection optimisation
late payers separated from real defaulters

Credit risk model enhancement
relational (graph) features added

Dynamic client limit
limit distributed across loan products

Risk analytics
early signs of repayment difficulty

TYPICAL DATA

«AX DOĞRULASIN»

- Applications, limits, payments and delinquency history
- Transactions, balances and channel interactions
- Collection actions and contact results
- Relations between clients: transfers, partners, guarantors

HOW SUCCESS IS MEASURED

Technical metric
Gini / AUC on hold-out data; explainable drivers

Business metric
Transition into delinquency; approval rate at the same NPL ratio

REFERENCES AND RESULTS

«AX DOĞRULASIN»

Albaraka Banking Group, TEB · collection optimisation
3–7% reduction in transition into the first delinquency cycle; Gini 0.86; contact policy optimised

QNB Finansbank · credit risk model enhancement
+4.5% credit approval rate at the same NPL ratio; Gini 0.78–0.81; graph analytics

Family 5: Industrial Operations

KOMTAS

Machine learning models that predict the process and guide settings, quality checks and maintenance within the plant's operating limits.

08 · INDUSTRIAL OPERATIONS

THE QUESTION

Which setting gives the lowest energy or the best yield?

Models that predict temperature, energy, quality or yield, and recommend set points within the plant's limits.

USE CASES IN THIS FAMILY

Energy optimisation
HVAC systems and furnaces

Predictive quality
defect classification, automated checks

Production yield
parameters that lower the yield

Predictive maintenance
failures predicted from IoT data

Process unit prediction
plant variables from lab and sensor data

Fixed asset utilisation
use of machines and production lines

TYPICAL DATA «AX DOĞRULASIN»

- Sensor and IoT time series (temperature, humidity, energy)
- Process parameters, controllable and uncontrollable
- Lab samples and quality records
- Production, yield and maintenance records

HOW SUCCESS IS MEASURED

Technical metric
Forecast error and classification accuracy on hold-out data

Business metric
Energy use, yield or deviation, test vs control period

REFERENCES AND RESULTS «AX DOĞRULASIN»

Eczacıbaşı Monrol · energy optimisation in HVAC
2.3% lower energy consumption in test units vs the control group

Şişecam · furnace energy and temperature, glass quality, yield
Result: «RESULT»

An integrated oil / petrochemical group · process unit prediction
More stable reformer operation (qualitative)

Results from the Field

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Results from Komtaş predictive analytics projects, by use-case family.

REFERENCES AND MEASURED RESULTS

«AX DOĞRULASIN»

Family	Use case	Reference	Result
Retention & Churn	Asset churn prediction	**Odeabank**	11% lower segment downgrade in the test group vs the control group
Demand & Supply	Demand prediction	**A tyre manufacturer**	7% lower stock holding cost for 7 tyre types
Demand & Supply	Visitor and sales prediction	**Shopping mall management**	Forecasts for 350+ stores in 2 malls, 1.6M visitors
Pricing & Personalization	List price and markdown optimisation	**A spare-parts seller**	+4% revenue at the same conversion in the test group; price elasticity for 30,000+ SKUs; 12 A/B test scenarios
Risk & Collections	Collection optimisation	**Albaraka Banking Group, TEB**	3–7% reduction in transition into the first delinquency cycle; Gini 0.86; explainable AI; contact policy optimised
Risk & Collections	Credit risk model enhancement	**QNB Finansbank**	+4.5% credit approval rate at the same NPL ratio; Gini 0.78–0.81; graph analytics
Industrial Operations	Energy optimisation in HVAC	**Eczacıbaşı Monrol**	2.3% lower energy consumption in test units vs the control group
Industrial Operations	Furnace energy and temperature, glass quality, yield	**Şişecam**	«RESULT»
Industrial Operations	Process unit prediction	**An integrated oil / petrochemical group**	More stable reformer operation (qualitative)

FURTHER FIELD EXPERIENCE

«GÜNER DOĞRULASIN»

Retention & Churn	Subscription churn	**A fitness / subscription company**	«RESULT»
Demand & Supply	Demand forecasting	**A retail company**	«RESULT»

Customer names are shown with the customers' permission. Unnamed references are shown by sector.

How Business Value Is Measured

KOMTAS

A good technical metric does not show that the business has gained. The effect is measured by comparing a group that uses the model with a group that does not.

09 · VALUE MEASUREMENT

CONTROL GROUP DESIGN · AGREED IN THE PILOT

Customers, products, stores or units

split at random

Target group
the model output is used: offer, price, order, contact, set point

Control group
the current way of working continues

Compare
after «NUMBER» weeks

Difference

the measured value of the model

BUSINESS METRIC PER FAMILY

«AX DOĞRULASIN»

- **Retention & Churn** — retention or downgrade difference vs the control group
- **Demand & Supply** — back-test vs the current method; stock cost, stock-outs
- **Pricing & Personalization** — A/B test: revenue, conversion, click-through rate
- **Risk & Collections** — delinquency transition; approval rate at the same NPL
- **Industrial Operations** — energy, yield or deviation vs a control period

TWO KINDS OF METRIC

«AX DOĞRULASIN»

Technical metric
Measured on hold-out data. Thresholds «NUMBER» are agreed before the work starts and form the acceptance.

Business metric
Measured in the control group test and reported. It depends on how the business uses the output, so it is not guaranteed.

The business must act on the output. If scores, prices or forecasts are not used in the business process, the value cannot be measured. This is agreed with the business owner in the Pilot.

Three Service Tiers: Pilot, Production, Run

KOMTAS

The Pilot proves one use case on the customer's data. Production puts the model into daily use. Run keeps it accurate month by month.

10 · SERVICE TIERS

A Pilot

fixed fee · one use case · 6–8 weeks

- Data readiness check
- Feature set and model
- Validation on hold-out data
- Value measurement plan and first results
- Go / no-go decision for Production

OUTPUT

Readiness report · model and feature definitions · validation report · Production plan and estimate

«AX DOĞRULASIN»

B Production

price per use case · MLOps foundation once

- MLOps foundation on the agreed platform
- Deployment pipeline and scheduled scoring
- Model registry and monitoring
- Integration to target systems (each a line item)
- Runbook and handover

OUTPUT

Model running on schedule · output in business systems · runbook handed over

«AX DOĞRULASIN»

C Run

monthly · minimum term 12 months

- Drift and performance monitoring
- Retraining on a schedule
- Monthly report
- Small improvements from monthly capacity
- Unused hours are not carried over

OUTPUT

Monthly report · models kept accurate · changes from a monthly capacity

«AX DOĞRULASIN»

Pilot first. Production starts from a validated model: the Pilot, or an equivalent model already validated by the customer. «AX DOĞRULASIN»

Pilot: Work Flow and Timeline

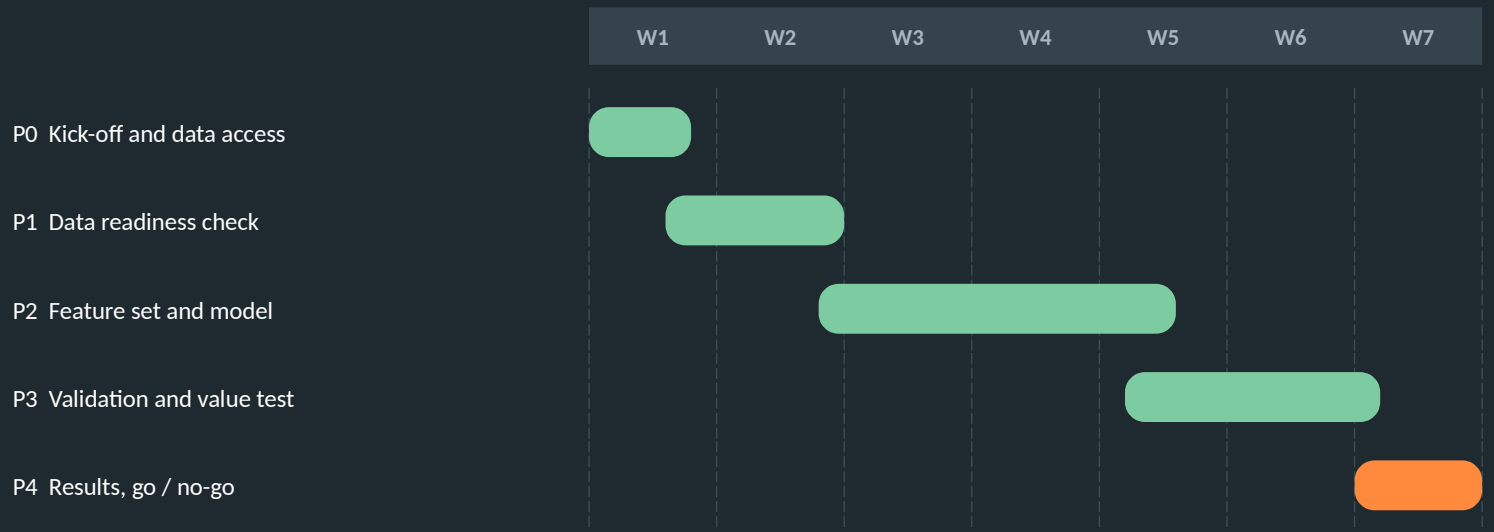
KOMTAS

Five phases over six to eight weeks, for one use case. The Pilot ends with results and a go / no-go decision for Production.

11 · PILOT

INDICATIVE PLAN · BAND M · 7 WEEKS

«AX DOĞRULASIN»



Go / no-go for Production. Decided on the validation results against the agreed threshold «NUMBER» and on the first value measurement. «AX DOĞRULASIN»

Duration by size. Small about 6, medium about 7, large about 8 weeks. Size depends on data sources, data volume, prediction frequency and integrations. «AX DOĞRULASIN»

DELIVERABLES «AX DOĞRULASIN»

- Data readiness report
- Model and feature definitions
- Validation report
- Value measurement plan and first results
- Production plan and estimate

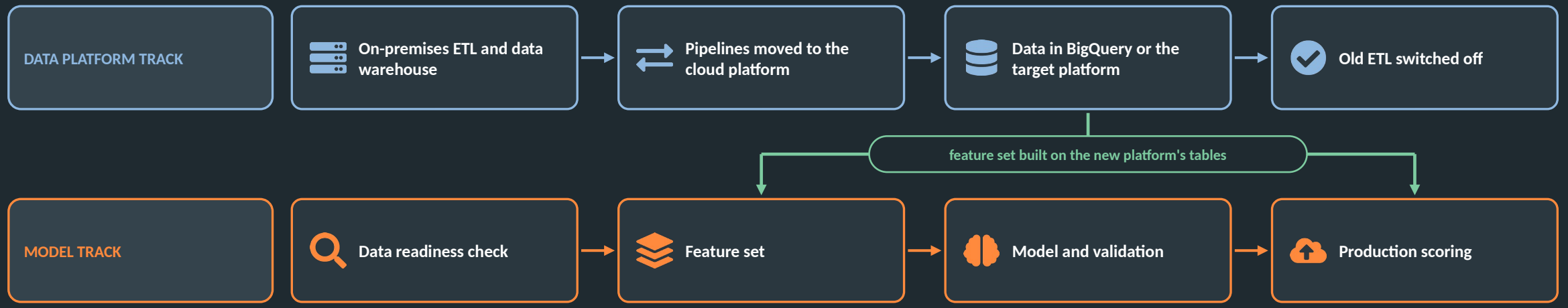
Data Platform Modernization and the Model

KOMTAS

When the data still sits in on-premises ETL and a data warehouse, the move to a cloud data platform and the model work can be planned together.

12 · DATA PLATFORM AND MODEL

«GÜNER DOĞRULASIN»



THREE WAYS TO PLAN IT

«AX DOĞRULASIN»

1 Model first

The Pilot runs on an extract of the current data. Production waits for the platform.

2 Platform first

The data moves first; the model is built on the new platform.

3 Together

The tables the first use case needs move first; the Pilot and the first migration wave run in parallel.

Why plan them together. Sources are moved in the order the model needs them, pipelines are not built twice, and one data team works to one plan.

About Komtaş, and the Next Step

Who runs Predictive Analytics, and what is needed to start.

KOMTAS

13 · ABOUT KOMTAŞ AND NEXT STEP

WHY KOMTAŞ

Google Cloud Premier partner
Komtaş is a Google Cloud Premier partner.

Microsoft Azure partner
Komtaş is also a Microsoft Azure partner.

Models in use in the field
Banking, manufacturing, retail and services; results on p. 10. «AX DOĞRULASIN»

MLOps foundation experience
MLOps platforms and model standardisation for banks and manufacturers. «AX DOĞRULASIN»

From data to models in use
Data platform, Pilot, Production and monthly Run from one partner.

Since 1989, focused on data · 250+ specialists · offices in Istanbul, Ankara, Izmir and Dubai

NEXT STEPS

1. **Walkthrough** — today — you are here
2. **Discovery call** — use case, business goal, data, platform
3. **Pilot proposal** — fixed fee, one use case
4. **Kick-off** — data access, readiness check starts

Needed from the customer. A business owner and the choice of the first use case; a data owner who gives access to the history; approval for the use of customer data under KVKK; IT for integration; agreement on a control group. The Pilot clock starts when data access is given. «AX DOĞRULASIN»